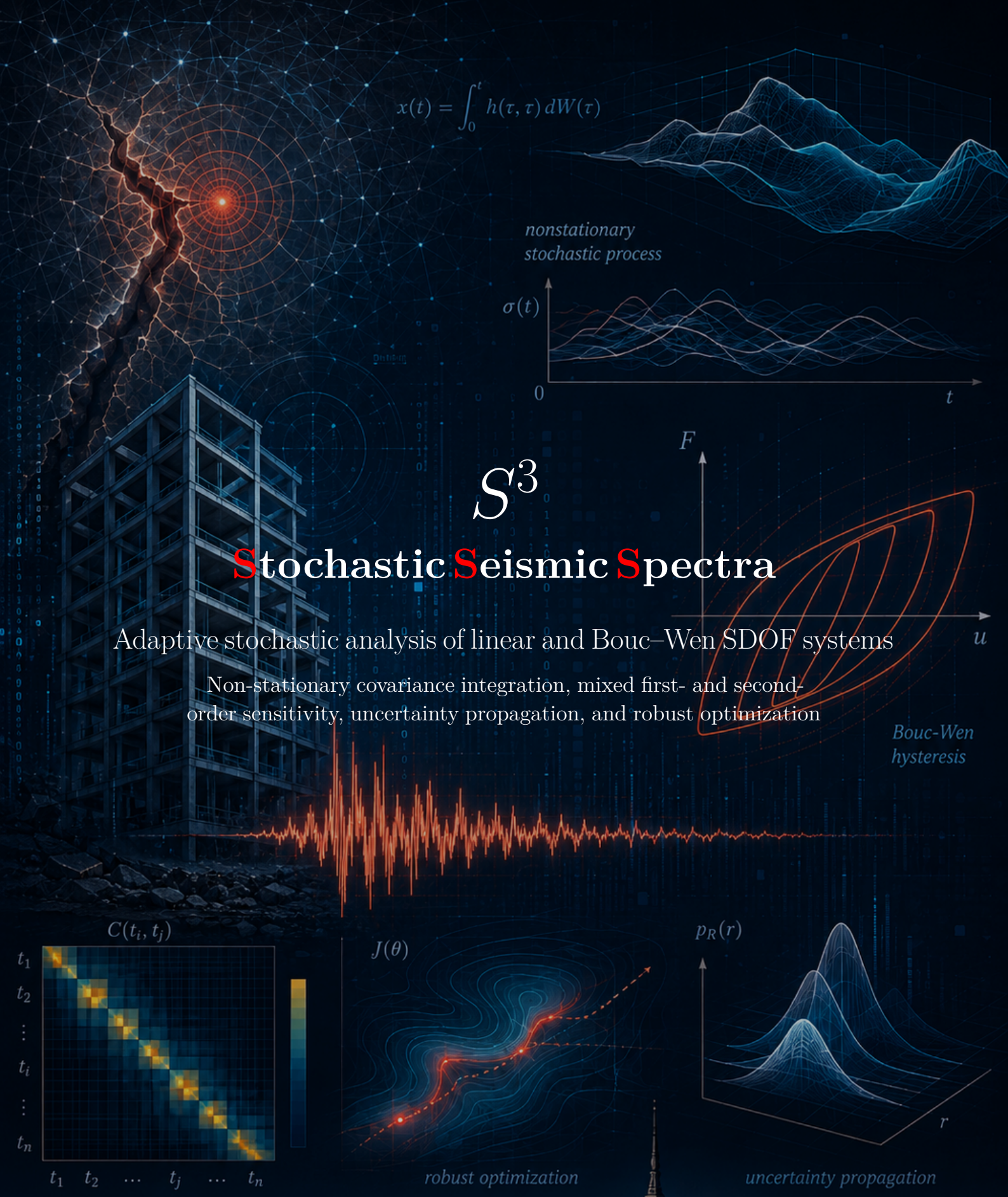


S^3

Stochastic Seismic Spectra

Adaptive stochastic analysis of linear and Bouc–Wen SDOF systems

Non-stationary covariance integration, mixed first- and second-order sensitivity, uncertainty propagation, and robust optimization

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Technical Lecture Notes and Methodological Report — Version 2.3-beta — October 2026

Abstract

SSS (Stochastic Seismic Spectra) is a research-oriented MATLAB toolbox for the time-domain random-vibration analysis of linear and equivalent-linearised Bouc–Wen single-degree-of-freedom systems under filtered, amplitude-modulated, and optionally frequency-modulated Gaussian excitation. The state covariance is governed by a non-stationary differential Lyapunov equation and is advanced on an automatically selected non-uniform grid by either an adaptive implicit trapezoidal method or an L-stable TR-BDF2 method. The software returns state covariance histories, the physical covariance of relative displacement, velocity, and absolute acceleration, marginal probability densities, first-passage peak estimates, and mean energy histories.

A central feature is the separation of two distinct stochastic levels: randomness of the excitation and uncertainty of the model or excitation parameters. Parametric uncertainty is propagated by a mixed-order strategy. First derivatives of the covariance are always used to quantify local sensitivity and the leading-order output variance; second derivatives are additionally used to correct the output mean and to include curvature contributions in the output covariance. Thus, the second-order calculation does not replace the first-order sensitivity: it augments it. For linear systems, the nominal covariance, gradient, and Hessian satisfy coupled differential Lyapunov equations and are integrated on the same adaptive grid. For the equivalent-linearised Bouc–Wen system, the current implementation differentiates the complete adaptive solver by centred finite differences. The final appendices present the full mixed-order derivation and a robust NSGA-II-oriented optimization framework in which expected performance and performance dispersion are treated as competing objectives.

Keywords: non-stationary random vibration; differential Lyapunov equation; adaptive time integration; covariance sensitivity; second-order uncertainty propagation; Bouc–Wen model; robust optimization; NSGA-II; MATLAB.

Table 1: Document control.

Document	SSS — Stochastic Seismic Spectra: Technical Lecture Notes and Methodological Report
Version	2.3-beta
Language	English
Primary software entry points	`sss_run_case`, `sss_run_uncertain_case`, `sss_config`
Numerical core	Adaptive trapezoidal and L-stable TR-BDF2 evolutionary-Lyapunov integration
Uncertainty method	Mixed first-/second-order perturbation: FOSM plus Gaussian SOSM corrections
Optimization method	Robust multiobjective formulation; NSGA-II / MATLAB `gamultiobj` implementation route
Suggested citation	G. C. Marano, L. Sardone, A. Fiore, R. Greco, G. Datta, A. Hajiha (2026). *SSS — Stochastic Seismic Spectra: Technical Lecture Notes and Methodological Report*, Version 2.3-beta. Zenodo (DOI: 10.5281/zenodo.23338165).
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Nomenclature

Symbol	Definition
$\mathbf{Z}(t)$	Augmented structural-filter state vector.
$\mathbf{A}(t)$	State matrix; possibly time-dependent.
$\mathbf{B}(t)$	White-noise intensity matrix in the covariance equation.
$\mathbf{R}(t)$	State covariance $\mathbb{E}[\mathbf{Z}\mathbf{Z}^{\mathsf{T}}]$.
$\mathbf{C}_y(t)$	Physical covariance of $\mathbf{y} = [x, v, a_{\text{abs}}]^{\mathsf{T}}$.
$\boldsymbol{\theta}$	Vector of uncertain parameters.
$\boldsymbol{\Sigma}_\theta$	Covariance matrix of uncertain parameters.
$\mathbf{R}_{,i}$	First parameter derivative $\partial\mathbf{R}/\partial\theta_i$.
$\mathbf{R}_{,ij}$	Second parameter derivative $\partial^2\mathbf{R}/(\partial\theta_i\partial\theta_j)$.
$\mathbf{g}_y$	Gradient of scalar output y with respect to $\boldsymbol{\theta}$.
$\mathbf{H}_y$	Hessian of scalar output y with respect to $\boldsymbol{\theta}$.
FOSM	First-order second-moment approximation.
SOSM	Second-order second-moment approximation.
aTR	Adaptive implicit trapezoidal integrator with step doubling.
TR-BDF2	Two-stage, second-order, L-stable implicit integrator.
P_f, r	Probability of failure and reliability, respectively. Note that the code input `P.Pf` is the target *non-exceedance* probability (a reliability level, e.g. 0.95), not a failure probability.
J	Generic scalar performance measure used in robust optimization.
$\mathbf{b}$	Deterministic design vector.
$\mathbf{d}$	Uncertain parameter vector in the optimization formulation.

1 Introduction and scope

Random-vibration analysis distinguishes the intrinsic randomness of a dynamic action from uncertainty in the parameters used to represent the structure and the action. If the parameters are fixed, the covariance matrix is a deterministic function of time even though the response is random. If masses, stiffnesses, damping ratios, filter frequencies, intensity parameters, or hysteretic parameters are uncertain, the covariance itself becomes an uncertain matrix-valued output. These two levels must not be conflated.

The first level is addressed through the state-space covariance equation. For Gaussian excitation and a linear state equation, the complete second-order response is obtained from the evolutionary Lyapunov equation. The second level is addressed through parameter derivatives of that covariance equation and moment-based uncertainty propagation. The original robust-design formulation of Marano et al. [1] used first derivatives of stationary covariance to determine the mean and variance of a performance measure. The non-stationary sensitivity manuscript [2] extended the direct differentiation of the Lyapunov equation to evolutionary covariance and reliability. The present software combines those ideas with variable-step, stability-controlled integration [3] and adds a Gaussian second-order moment correction.

The term *second order* is used in two different senses and requires explicit clarification. The covariance $\mathbf{R}$ is a second statistical moment of the dynamic response. By contrast, first- or second-order *sensitivity* refers to the derivative order with respect to uncertain parameters. SSS uses the following mixed strategy:

(i) the first derivatives $\mathbf{R}_{,i}$ are always the primary sensitivity quantities and always enter the leading-order variance of every uncertain output;

(ii) the second derivatives $\mathbf{R}_{,ij}$ are additionally calculated when `P.unc.order = 2`; they correct the output mean and add a curvature term to the output covariance;

(iii) the second-order result therefore uses both the gradient and the Hessian, rather than only the Hessian.

The software is intended for research, verification, parametric studies, spectrum generation, and robust optimization. It is not a substitute for model validation, careful probabilistic characterisation, or nonlinear Monte Carlo verification when uncertainty is large.

1.1 Main capabilities

The current SSS release provides: a linear SDOF model; a non-degrading Bouc–Wen model with $n = 1$ and equivalent Gaussian linearisation; a Clough–Penzien ground-motion filter with a Jennings-type envelope; constant or time-dependent filter parameters; adaptive covariance integration; covariance gradients and Hessians; correlated parameter uncertainty; confidence bands; probability-density grids; first-passage peaks; energy histories; and a software structure suitable for embedding in a robust optimization loop.

2 Stochastic dynamic formulation

2.1 Linear structural oscillator

For a linear SDOF system of circular frequency ω_0 and viscous damping ratio ξ, the relative displacement $x(t)$ satisfies

$$\ddot{x}(t) + 2\xi\omega_0\dot{x}(t) + \omega_0^2 x(t) = -\ddot{x}_g(t). \tag{1}$$

The ground acceleration is obtained from a filtered white-noise process. The implemented Clough–Penzien representation combines a Kanai–Tajimi component (ω_g, ξ_g) with a high-pass

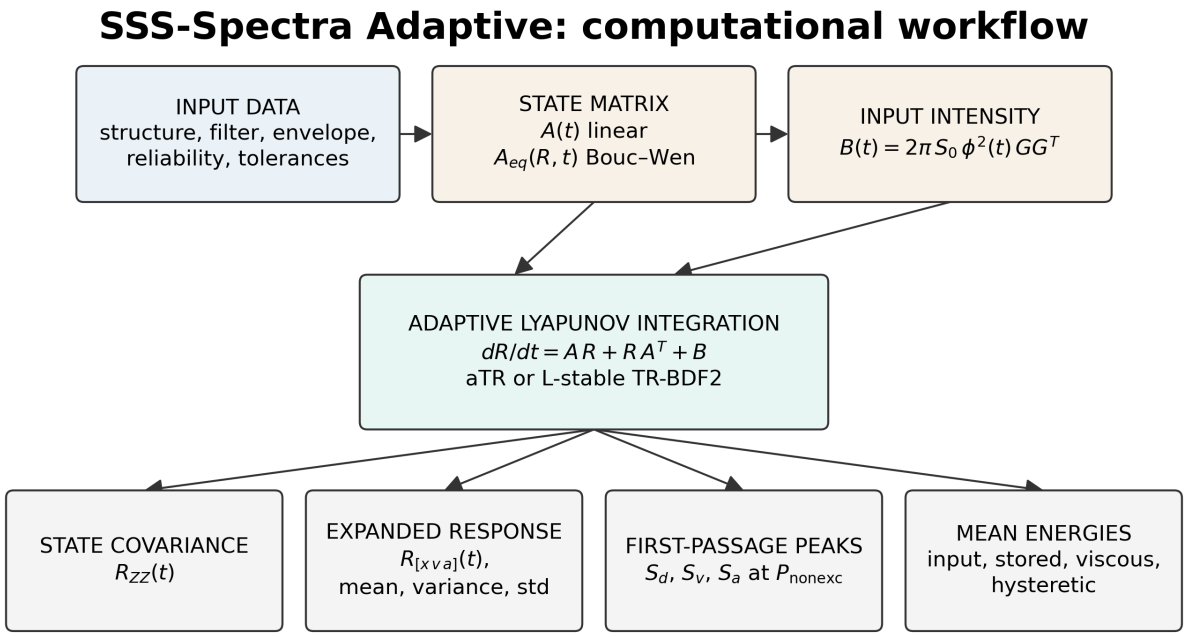


Figure 1: Overall computational workflow of SSS (Stochastic Seismic Spectra).

component (ω_p, ξ_p) [4–6]. A deterministic envelope $\phi(t)$ produces non-stationary intensity, while optional functions `wg_fun`, `xg_fun`, `wp_fun`, and `xp_fun` permit deterministic frequency or damping modulation.

The six-dimensional linear state used by the code is

$$\mathbf{Z} = \begin{bmatrix} x_s & x_p & x_g & v_s & v_p & v_g \end{bmatrix}^{\mathsf{T}}, \tag{2}$$

with state equation

$$\dot{\mathbf{Z}}(t) = \mathbf{A}(t)\mathbf{Z}(t) + \mathbf{G}W(t). \tag{3}$$

Here $W(t)$ is a zero-mean Gaussian white noise. The state matrix is assembled by `sss_Alin.m`.

2.2 Bouc–Wen oscillator and equivalent Gaussian linearisation

For the non-degrading Bouc–Wen oscillator, the restoring force is separated into elastic and hysteretic parts. In the implemented normalisation,

$$\ddot{x} + 2\xi\omega_0\dot{x} + \alpha\omega_0^2 x + (1-\alpha)\omega_0^2 z = -\ddot{x}_g, \tag{4}$$

while the hysteretic internal variable satisfies

$$\dot{z} = A_z\dot{x} - \beta_{BW}|\dot{x}|\,|z|^{n-1}z - \gamma_{BW}\dot{x}|z|^n. \tag{5}$$

The current closed-form equivalent-linearisation implementation is restricted to $n = 1$. The nonlinear equation is replaced at every implicit stage by

$$\dot{z} = (A_z - c_e)\dot{x} - k_e z, \tag{6}$$

where

$$c_e = \sqrt{\frac{2}{\pi}}\,\sigma_z\left(\beta_{BW}\rho_{\dot{x}z} + \gamma_{BW}\right), \tag{7}$$

$$k_e = \sqrt{\frac{2}{\pi}}\,\sigma_{\dot{x}}\left(\beta_{BW} + \gamma_{BW}\rho_{\dot{x}z}\right). \tag{8}$$

These expressions follow the equivalent Gaussian linearisation framework for hysteretic systems [7–9]. The resulting state matrix $\mathbf{A}_{eq}(\mathbf{R}, t)$ depends on the covariance and the differential Lyapunov equation becomes nonlinear. The implicit stages are therefore completed by a fixed-point loop on (c_e, k_e) before the local-error test is performed.

2.3 Evolutionary covariance equation

For zero mean, the state covariance

$$\mathbf{R}(t) = \mathbb{E}\left[\mathbf{Z}(t)\mathbf{Z}^{\mathsf{T}}(t)\right] \tag{9}$$

obeys

$$\dot{\mathbf{R}}(t) = \mathbf{A}(t)\mathbf{R}(t) + \mathbf{R}(t)\mathbf{A}^{\mathsf{T}}(t) + \mathbf{B}(t), \qquad \mathbf{R}(0) = \mathbf{R}_0. \tag{10}$$

For the present filtered-white-noise arrangement,

$$\mathbf{B}(t) = 2\pi S_0 \phi^2(t)\mathbf{G}\mathbf{G}^{\mathsf{T}}. \tag{11}$$

The spectral level S_0 is calibrated from the target PGA and a peak factor. The envelope uses a quadratic rise, a plateau, and an exponential decay, consistent with the Jennings representation [10]. The same matrix equation supports constant $\mathbf{A}$, time-varying $\mathbf{A}(t)$, and the covariance-dependent $\mathbf{A}_{eq}(\mathbf{R}, t)$ of the Bouc–Wen equivalent-linearisation route.

3 Adaptive variable-step Lyapunov integration

Fixed-step covariance algorithms require the analyst to select a time increment that simultaneously resolves the fastest oscillatory mode, the corners of the modulation envelope, and the slowly decaying tail. SSS-Spectra Adaptive instead uses local error control and step rejection. Each accepted state is located on a non-uniform time grid.

3.1 Adaptive implicit trapezoidal method

For a trial step $h_n = t_{n+1} - t_n$, the variable-step trapezoidal update is written as the algebraic Lyapunov equation

$$\mathbf{P}_B\mathbf{R}_{n+1} + \mathbf{R}_{n+1}\mathbf{P}_B^{\mathsf{T}} = \mathbf{P}_F\mathbf{R}_n + \mathbf{R}_n\mathbf{P}_F^{\mathsf{T}} + \frac{h_n}{2}\left(\mathbf{B}_n + \mathbf{B}_{n+1}\right), \tag{12}$$

where

$$\mathbf{P}_B = \frac{1}{2}(\mathbf{I} - h_n\mathbf{A}_{n+1}), \qquad \mathbf{P}_F = \frac{1}{2}(\mathbf{I} + h_n\mathbf{A}_n). \tag{13}$$

One full step and two half steps provide a Richardson estimate. The extrapolated two-half-step value is accepted when the scaled root-mean-square error is not greater than one. The method is A-stable but not L-stable; an optional oscillatory cap limits the step relative to the largest imaginary part of the state-matrix eigenvalues.

3.2 L-stable TR-BDF2 method

TR-BDF2 combines a trapezoidal substep at $t_n + \gamma h_n$, $\gamma = 2 - \sqrt{2}$, and a BDF2 substep at t_{n+1} [11, 12]. Each substage is again an algebraic Lyapunov solve. An embedded estimate is formed without an additional solve. L-stability makes this method the preferred default for lightly damped systems and for covariance blocks containing stiff filter or equivalent-linearisation components.

3.3 PI step-size controller

The scaled error is

$$\mathrm{err} = \|\mathbf{E} \oslash (\mathrm{atol} + \mathrm{rtol}\max(|\mathbf{X}_n|, |\mathbf{X}_{n+1}|))\|_{\mathrm{RMS}}\,, \tag{14}$$

where $\mathbf{X}$ denotes the covariance alone or the stacked covariance-sensitivity state. Accepted steps satisfy $\mathrm{err} \le 1$. A PI controller updates the trial step using the current and previous errors, which reduces oscillatory step sequences near envelope corners [13]. Rejected steps are repeated with a smaller h.

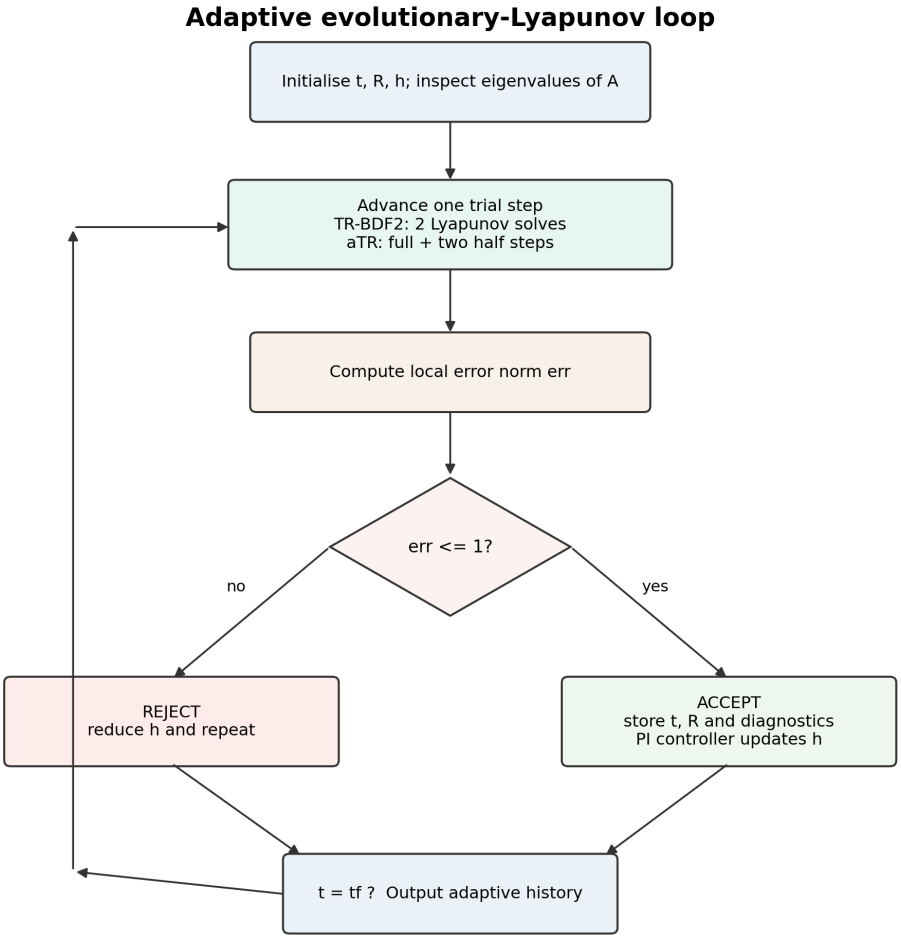


Figure 2: Adaptive evolutionary-Lyapunov integration loop.

3.4 Matrix structure and numerical safeguards

The algorithm operates directly on symmetric matrices. Each stage calls `sss_solve_lyap`; MATLAB's `lyap` is used when available, with a dense Kronecker fallback for the small matrices considered here. The accepted covariance is symmetrised. Small negative eigenvalues caused by round-off may be clipped if `P.num.enforce_psd = true`. Sensitivity matrices are symmetrised but never projected, because a derivative of a covariance matrix is not itself required to be positive semidefinite.

4 Physical outputs, probability densities, reliability, and energy

4.1 Expanded covariance of displacement, velocity, and acceleration

The physical output is

$$\mathbf{y}(t) = \begin{bmatrix} x(t) & v(t) & a_{\mathrm{abs}}(t) \end{bmatrix}^{\mathsf{T}} = \mathbf{H}(t)\mathbf{Z}(t), \tag{15}$$

with covariance

$$\mathbf{C}_y(t) = \mathbf{H}(t)\mathbf{R}(t)\mathbf{H}^{\mathsf{T}}(t). \tag{16}$$

The code returns the complete 3×3 matrix, including cross-covariances. Absolute acceleration is obtained from the structural equation of motion and is not the relative acceleration. The principal fields are `response.cov`, `response.var`, `response.std`, `response.max_std`, and `response.peak`.

4.2 Response probability densities

For fixed parameters, the linear response is Gaussian and zero mean. The marginal density of component y_q is

$$p_{y_q}(y,t) = \frac{1}{\sqrt{2\pi C_{qq}(t)}} \exp\left[-\frac{y^2}{2C_{qq}(t)}\right]. \tag{17}$$

For the equivalent-linearised Bouc–Wen model, the same Gaussian representation is an approximation consistent with the closure used to calculate c_e and k_e. Under parametric uncertainty, the toolbox provides moment-matched density grids for the covariance entries and uses the propagated mean physical covariance for joint Gaussian response-density evaluation.

4.3 First-passage peaks

Reliability-based peak estimates use the covariance of a process, the covariance of its time derivative, and their cross-covariance. A Vanmarcke/Rice-type crossing approximation is integrated on the same adaptive grid [14]. The input `P.Pf` is retained for legacy compatibility but represents a non-exceedance probability. The resulting ordinates `Sd`, `Sv`, and `Sa` are probabilistic peaks, not deterministic maxima of one time history.

4.4 Mean energy histories

For the linear system the package calculates mean kinetic, elastic, viscous-dissipation, and input-energy histories from covariance terms. For the Bouc–Wen model it additionally calculates hysteretic energy. These quantities can be assembled into energy spectra over the period grid `P.T`.

5 Mixed-order sensitivity and uncertainty strategy

This section gives the operational interpretation. The complete derivation is provided in Appendix A.

Let $\boldsymbol{\theta} = [\theta_1, \ldots, \theta_{n_p}]^{\mathsf{T}}$ be the uncertain-parameter vector with mean $\bar{\boldsymbol{\theta}}$ and covariance $\boldsymbol{\Sigma}_\theta$. The nominal physical and stochastic data stored in `P` are interpreted as parameter means. The user may provide a full covariance matrix, a vector of standard deviations, or coefficients of variation.

5.1 First derivatives: the primary sensitivity

For each parameter, the first covariance sensitivity is

$$\mathbf{R}_{,i} = \frac{\partial \mathbf{R}}{\partial \theta_i}. \tag{18}$$

It satisfies another differential Lyapunov equation with the same homogeneous operator as the nominal covariance. These derivatives quantify whether an increase of a parameter increases or

decreases each response covariance and determine the leading-order dispersion caused by parameter uncertainty.

5.2 Second derivatives: curvature correction

When `P.unc.order = 2`, the Hessian blocks

$$\mathbf{R}_{,ij} = \frac{\partial^2 \mathbf{R}}{\partial\theta_i \partial\theta_j} \tag{19}$$

are also evaluated. They represent local curvature of the covariance with respect to the uncertain parameters. They are used to correct the output mean and to add a quadratic contribution to output covariance. They do not replace $\mathbf{R}_{,i}$.

5.3 FOSM and SOSM outputs

For a scalar output $y(\boldsymbol{\theta})$ with gradient $\mathbf{g}_y$ and Hessian $\mathbf{H}_y$, the first-order result is

$$\mathbb{E}[y]_{\mathrm{FOSM}} \approx y_0, \qquad \mathrm{Var}[y]_{\mathrm{FOSM}} \approx \mathbf{g}_y^{\mathsf{T}} \boldsymbol{\Sigma}_\theta \mathbf{g}_y. \tag{20}$$

For zero-mean Gaussian parameter perturbations, the second-order result is

$$\mathbb{E}[y]_{\mathrm{SOSM}} \approx y_0 + \frac{1}{2}\,\mathrm{tr}(\mathbf{H}_y \boldsymbol{\Sigma}_\theta), \tag{21}$$

$$\mathrm{Cov}[y,z]_{\mathrm{SOSM}} \approx \mathbf{g}_y^{\mathsf{T}} \boldsymbol{\Sigma}_\theta \mathbf{g}_z + \frac{1}{2}\,\mathrm{tr}(\mathbf{H}_y \boldsymbol{\Sigma}_\theta \mathbf{H}_z \boldsymbol{\Sigma}_\theta). \tag{22}$$

The first term of the SOSM covariance is the first-order gradient term. Consequently, the implemented method is correctly described as a *mixed first-/second-order approach*.

6 Software architecture

The package separates physical modelling, numerical integration, uncertainty propagation, post-processing, and plotting. This separation permits validation of each layer and facilitates replacement of the SDOF model by an MDOF or finite-element state matrix.

Table 2: Principal MATLAB functions.

Function	Purpose
`sss_config.m`	Central physical, numerical, reliability, spectrum, and uncertainty inputs.
`sss_run_case.m`	Complete nominal linear or Bouc–Wen analysis.
`sss_run_uncertain_case.m`	Nominal covariance, first and second sensitivities, FOSM/-SOSM moments, bands, and PDFs.
`sss_lyap_adaptive.m`	Adaptive nominal evolutionary-Lyapunov integration.
`sss_lyap_adaptive_sens.m`	Joint covariance and first-sensitivity integration.
`sss_lyap_adaptive_sens2.m`	Joint covariance, gradient, and Hessian integration.
`sss_parameter_derivatives_linear.m`	Linear-model derivatives of $\mathbf{A}$ and $\mathbf{B}$.
`sss_finite_difference_cov_sens.m`	Complete-solver centred differences for Bouc–Wen sensitivities.
`sss_propagate_matrix_uncertainty.m`	FOSM and Gaussian SOSM propagation of matrix outputs.
`sss_probability_densities.m`	Marginal response and covariance-entry density grids.

Function	Purpose
`sss_response_stats.m`	Transformation to $[x, v, a_{\mathrm{abs}}]$ covariance and peak statistics.
`sss_energy_history.m`	Mean energy histories.
`sss_selftest.m`	Reduced regression and consistency checks.

7 Installation and first execution

Extract the package and add it recursively to the MATLAB path:

```
addpath(genpath('SSS_Spectra_Adaptive_v2_1'));
report = sss_selftest();
```

MATLAB R2018b or later is recommended. The Control System Toolbox is useful because it provides `lyap`, but a fallback is included. Before a publication campaign, run the self-test after installation and after every modification of the numerical core.

A nominal linear and Bouc–Wen analysis is launched as follows:

```
P = sss_config();
L = sss_run_case(P,'linear',0.50);
B = sss_run_case(P,'boucwen',0.50);
```

8 Input data

8.1 Physical and stochastic input

The central structure `P` is generated by `sss_config`. Table 3 lists the principal groups.

Table 3: Principal input groups.

Group	Contents
`P.env`	Envelope rise time, plateau end, and exponential decay coefficient.
`P.ground`	Filter frequencies and damping ratios, PGA, peak factor, optional time laws, and calibrated S_0.
`P.lin`	Linear SDOF damping ratio. The natural period is supplied to the run function.
`P.bw`	Bouc–Wen damping, post-yield stiffness ratio, yield displacement, A_z, n, β_{BW}, and γ_{BW}.
`P.num`	Integrator, tolerances, step limits, PI-controller settings, stability and PSD controls, and fixed-point settings.
`P.Pf`	Target non-exceedance probability used for first-passage peaks.
`P.T`	Period grid for spectral studies.
`P.unc`	Uncertain-parameter names, covariance information, derivative order, confidence level, Hessian mode, and finite-difference settings.

8.2 Uncertain parameters

Supported names include `wg`, `xg`, `wp`, `xp`, `PGA`, `peakf`, `S0`, `xi`, `lin_xi`, `bw_xi`, `bw_alpha`, `bw_uy`, `bw_A`, `bw_beta`, and `bw_gamma`. The nominal values are read from `P`.

Independent uncertainty may be entered through coefficients of variation:

```
P.unc.names = {'wg','xg','PGA','xi'};
P.unc.cov = [0.10 0.15 0.08 0.20];
```

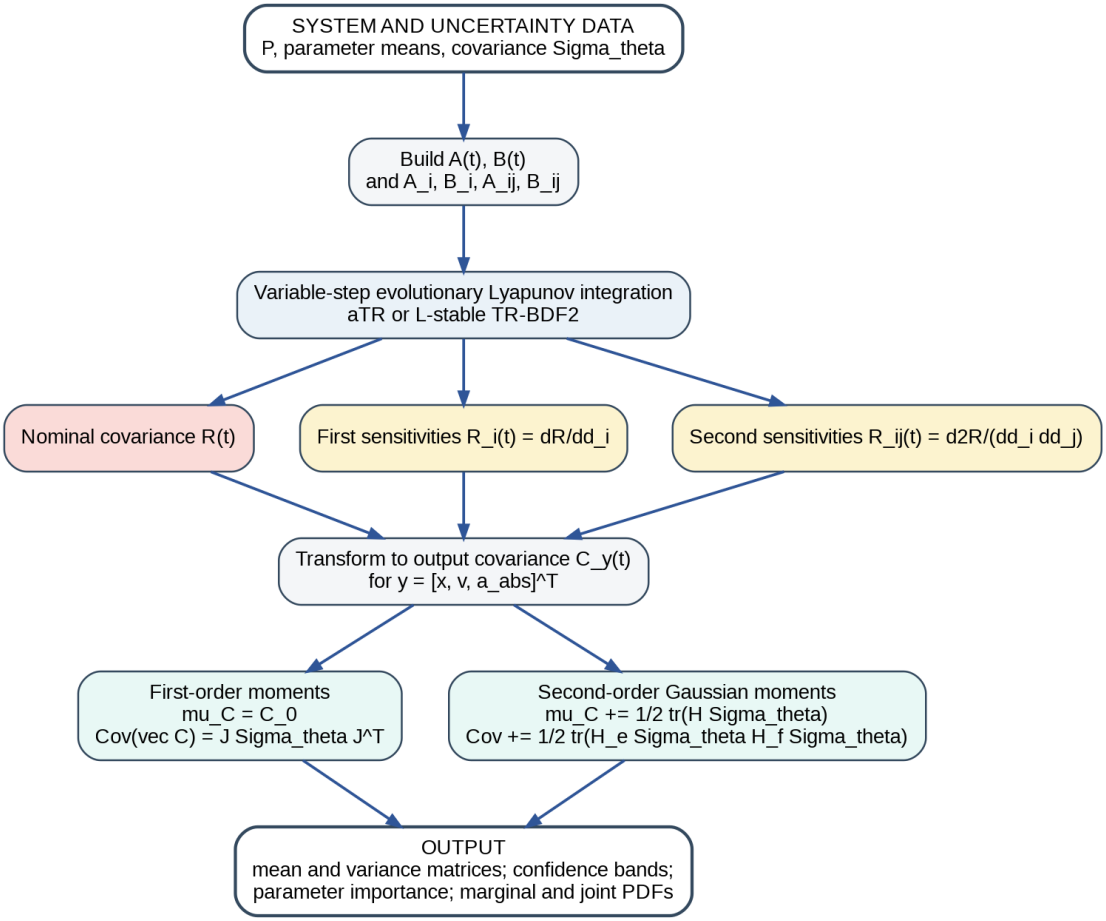


Figure 3: Mixed-order covariance-sensitivity and uncertainty-propagation workflow.

or standard deviations:

```
P.unc.std = [1.5 0.06 0.25 0.01];
```

Note the distinction: `P.unc.cov` is a *vector of coefficients of variation*, whereas `P.unc.covariance` (below) is a full parameter *covariance matrix*. A full covariance matrix has precedence and permits correlation:

```
mu = [P.ground.wg; P.ground.xg; P.ground.PGA; P.lin.xi];
cv = [0.10;0.15;0.08;0.20];
s = cv.*abs(mu);
Corr = eye(4);
Corr(1,2) = -0.35; Corr(2,1) = -0.35;
P.unc.covariance = diag(s)*Corr*diag(s);
```

The matrix is symmetrised and checked for positive semidefiniteness.

8.3 Mixed-order controls

The recommended specification is

```
P.unc.order = 2; % 1: FOSM only; 2: mixed FOSM+SOSM
P.unc.second_order_mode = 'full'; % or 'diagonal'
P.unc.distribution = 'gaussian'; % required by SOSM variance closure
P.unc.confidence = 0.95;
```

With `order = 1`, only nominal values and gradients are used. With `order = 2`, the first-order arrays remain present and the Hessian is added. The `diagonal` Hessian mode includes $\mathbf{R}_{,ii}$ only and omits mixed derivatives; it is useful for preliminary screening but is not equivalent to the full second-order expansion when parameters interact.

9 Running nominal and uncertain analyses

9.1 Nominal linear case

```
P = sss_config();
P.analysis.period = 0.50;
OUT = sss_run_case(P,'linear',P.analysis.period);
```

9.2 Nominal Bouc–Wen case

```
P = sss_config();
P.bw.alpha = 0.05;
P.bw.uy = 0.02;
OUT = sss_run_case(P,'boucwen',0.50);
```

9.3 Mixed-order uncertain linear case

```
P = sss_config();
P.unc.names = {'wg','xg','wp','xp','PGA','xi'};
P.unc.cov = [0.10 0.10 0.10 0.10 0.08 0.15];
P.unc.order = 2;
P.unc.second_order_mode = 'full';
P.unc.derivative_method = 'direct';
P.unc.confidence = 0.95;

```

```
OUT = sss_run_uncertain_case(P,'linear',0.50);
sss_plot_uncertain_case(OUT);
```

For the linear model, the covariance, first derivatives, and second derivatives are integrated jointly on the same adaptive grid.

9.4 Mixed-order uncertain Bouc–Wen case

```
P = sss_config();
P.unc.names = {'wg','PGA','bw_xi','bw_alpha','bw_uy'};
P.unc.cov = [0.10 0.08 0.20 0.15 0.10];
P.unc.order = 2;
P.unc.second_order_mode = 'diagonal';
P.unc.derivative_method = 'finite_difference';

OUT = sss_run_uncertain_case(P,'boucwen',0.50);
```

A full Bouc–Wen Hessian is possible but expensive: every mixed derivative requires several complete adaptive analyses. The diagonal option is appropriate for screening; the full option should be restricted to the influential parameters identified by the first-order analysis.

10 Output data model

The top-level output returned by `sss_run_uncertain_case` is organised as shown in Table 4.

Table 4: Principal uncertain-analysis outputs.

Field	Meaning
`OUT.nominal.state_covariance`	Nominal state covariance history.
`OUT.nominal.response.cov`	Nominal 3×3 covariance of $[x, v, a_{\mathrm{abs}}]$.
`OUT.sensitivity.state_covariance.first`	$\mathbf{R}_{,i}$ arrays.
`OUT.sensitivity.state_covariance.second`	$\mathbf{R}_{,ij}$ arrays.
`OUT.sensitivity.response_covariance.first`	First derivatives of $\mathbf{C}_y$.
`OUT.sensitivity.response_covariance.second`	Second derivatives of $\mathbf{C}_y$.
`OUT.uncertainty.response_covariance.first_order`	FOSM mean, covariance, standard deviation, bands, and importance.
`OUT.uncertainty.response_covariance.second_order`	Mixed-order SOSM mean, covariance, standard deviation, and bands.
`OUT.probability_density`	Marginal response and covariance-entry density grids.
`OUT.parameters`	Means, standard deviations, covariance, correlation, and differentiation steps.
`OUT.integration`	Accepted/rejected steps, local errors, step sizes, PSD projections, and stability diagnostics.

The array dimensions are important. If the state dimension is n_z, the number of accepted

time points is n_t, and the number of uncertain parameters is n_p,

$$\mathbf{R}: \; n_z \times n_z \times n_t, \tag{23}$$

$$\mathbf{R}_{,i}: \; n_z \times n_z \times n_t \times n_p, \tag{24}$$

$$\mathbf{R}_{,ij}: \; n_z \times n_z \times n_t \times n_p \times n_p. \tag{25}$$

The complete covariance between the independent upper-triangular entries of a matrix output is stored in `output_covariance(:,:,k)`. This is different from the physical response covariance itself.

11 Confidence bands and probability-density interpretation

For a central confidence probability p, the element-wise band is

$$\mu_y \pm z_p \sigma_y, \qquad z_p = \Phi^{-1}\left(\frac{1+p}{2}\right). \tag{26}$$

At $p = 0.95$, $z_p \simeq 1.96$. Lower limits of diagonal covariance entries are clipped at zero. The assembled lower and upper matrices are not guaranteed to be jointly positive semidefinite, because the intervals are calculated entry by entry. The software therefore also returns a PSD-projected second-order mean for joint Gaussian PDF evaluation.

Two covariance concepts must remain separate:

(a) $\mathbf{C}_y(t)$ is the physical covariance of displacement, velocity, and acceleration caused by the random excitation at fixed parameter values;

(b) $\mathrm{Cov}[\mathrm{vech}(\mathbf{C}_y)]$ is the covariance among the calculated entries of $\mathbf{C}_y$ caused by uncertain parameters.

The second quantity has squared covariance units and measures epistemic or parametric dispersion of the computed covariance.

12 Numerical recommendations and verification

12.1 Adaptive integration tolerances

For routine engineering studies, `rtol` between 10^{-4} and 10^{-5} is a reasonable starting range. Second derivatives may converge more slowly than the nominal covariance. A publication-grade study should repeat the analysis with `rtol` divided by ten and compare the response quantities used in the conclusions.

12.2 Finite-difference steps

The Bouc–Wen route and the current linear matrix-derivative builder use centred differences where required. The default perturbation is

$$h_i = \max\left(h_{\mathrm{abs}}, h_{\mathrm{rel}} \max(|\bar{\theta}_i|, 1)\right), \tag{27}$$

with a positivity safeguard. Derivatives should be repeated with $0.5h_i$ and $2h_i$. Stable gradients and Hessian contractions are more important than agreement of every small Hessian entry.

12.3 Validation hierarchy

The recommended hierarchy is: run `sss_selftest`; verify adaptive convergence; check finite-difference step convergence; compare selected derivatives with complete-solver finite differences; and, when coefficients of variation are moderate or large, compare important propagated moments with Monte Carlo or Latin hypercube sampling. The local Taylor expansion should be treated cautiously when the SOSM mean correction is not small compared with the nominal value.

12.4 Computational cost

For the direct linear route, one covariance, n_p gradient matrices, and up to $n_p(n_p+1)/2$ Hessian matrices are advanced. All share the same Lyapunov operator at a stage. Storage scales with $n_z^2 n_t n_p^2$. For SDOF state dimensions this is modest, but full-Hessian campaigns with many uncertain parameters should use importance screening, diagonal Hessians, or reduced stored histories.

13 Worked data-extraction examples

The following code extracts the second-order mean and confidence band of the displacement variance C_{xx}:

```
S = OUT.uncertainty.response_covariance.second_order;
t = OUT.t;
mu_Cxx = squeeze(S.mean(1,1,:));
lo_Cxx = squeeze(S.lower(1,1,:));
up_Cxx = squeeze(S.upper(1,1,:));
plot(t,mu_Cxx,'LineWidth',1.5); hold on;
plot(t,lo_Cxx,'--',t,up_Cxx,'--');
```

An approximate visual band for the response standard deviation is obtained by applying the square root to non-negative variance bounds:

```
mu_sigx = sqrt(max(mu_Cxx,0));
lo_sigx = sqrt(max(lo_Cxx,0));
up_sigx = sqrt(max(up_Cxx,0));
```

For a formally second-order expansion of $\sigma_x = \sqrt{C_{xx}}$, its gradient and Hessian should instead be obtained by the chain rule described in Appendix A.

Parameter importance for the eth independent matrix entry at the kth time point is available as

```
I = OUT.uncertainty.response_covariance.first_order ...
      .normalized_importance(e,:,k);
```

With correlated inputs, signed contributions can be negative because covariance cancellation is possible. Their sum recovers the first-order variance, apart from machine-level PSD cleanup.

14 Limitations and intended extensions

The direct gradient/Hessian equations are currently implemented for the linear oscillator. The Bouc–Wen model uses complete-solver finite differences; an analytic differentiation of the equivalent-linearisation fixed point is a planned extension. The Bouc–Wen model is non-degrading and restricted to $n = 1$. For strongly hysteretic response, the Gaussian closure underlying the equivalent linearisation can misestimate the response variances, and peak estimates derived from a Gaussian response are least reliable for absolute acceleration, whose distribution is non-Gaussian when the restoring force saturates; verification against Monte Carlo simulation of the full Bouc–Wen equation is recommended when large ductility demands are expected. Confidence bands and covariance-entry PDFs use Gaussian moment matching. The SOSM covariance formula assumes Gaussian parameter perturbations; non-Gaussian cases require explicit third- and fourth-order central moments, polynomial chaos, stochastic collocation, or simulation. The first-passage module quantifies excitation-induced crossing for fixed parameters; complete mixed-order uncertainty propagation of failure probability requires differentiating the hazard integral and is a natural extension of the sensitivity formulation in Greco and Trentadue [2].

15 Summary of recommended practice

Use the adaptive TR-BDF2 method as the default. Begin with a nominal analysis, then calculate first-order sensitivities for all plausible uncertain parameters. Use the importance measures to remove negligible parameters. Activate full second-order derivatives for the reduced set, especially when the FOSM result is strongly sensitive to parameter values or when a non-negligible mean correction is expected. Validate the final robust-design candidates with tighter tolerances and an independent sampling method. In optimization, use first-order results for population-wide screening and second-order results for Pareto-front refinement unless the design space is small enough to support a full mixed-order evaluation at every candidate.

A Mixed first- and second-order covariance sensitivity and uncertainty propagation

This appendix is normative for the interpretation of all sensitivity and uncertainty fields returned by the toolbox. It also clarifies the relationship between the first-order formulations in the reference studies and the second-order extension used here.

A.1 Two meanings of "order"

The covariance $\mathbf{R} = \mathbb{E}[\mathbf{Z}\mathbf{Z}^\mathsf{T}]$ is a second statistical moment. This fact does not imply that its parameter sensitivity is second order. A first-order covariance sensitivity is the first derivative $\mathbf{R}_{,i}$. The robust TMD formulation of Marano et al. [1] and the non-stationary reliability-sensitivity formulation of Greco and Trentadue [2] are first-order derivative methods. Their output variance is calculated from the gradient and the covariance of the uncertain parameters.

The present extension retains that first-order layer and adds second derivatives $\mathbf{R}_{,ij}$. The resulting moment calculation is mixed order:

- first derivatives determine local sensitivity and the leading-order output covariance;
- second derivatives correct the mean and add the quadratic covariance term;
- both derivatives are required by the second-order result.

A.2 Parameter-dependent evolutionary Lyapunov equation

Let

$$\dot{\mathbf{R}}(\boldsymbol{\theta}, t) = \mathbf{A}(\boldsymbol{\theta}, t)\mathbf{R}(\boldsymbol{\theta}, t) + \mathbf{R}(\boldsymbol{\theta}, t)\mathbf{A}^\mathsf{T}(\boldsymbol{\theta}, t) + \mathbf{B}(\boldsymbol{\theta}, t). \tag{28}$$

The initial covariance may also depend on $\boldsymbol{\theta}$. The current examples use $\mathbf{R}_0 = \mathbf{0}$, hence all initial derivatives are zero.

Differentiating equation (28) with respect to θ_i gives

$$\dot{\mathbf{R}}_{,i} = \mathbf{A}\mathbf{R}_{,i} + \mathbf{R}_{,i}\mathbf{A}^\mathsf{T} + \mathbf{Q}_i, \tag{29}$$

where

$$\mathbf{Q}_i = \mathbf{A}_{,i}\mathbf{R} + \mathbf{R}\mathbf{A}_{,i}^\mathsf{T} + \mathbf{B}_{,i}. \tag{30}$$

The homogeneous operator is identical to the primal operator. At every stage the nominal covariance is solved first, $\mathbf{Q}_i$ is assembled, and each $\mathbf{R}_{,i}$ is obtained from an algebraic Lyapunov equation with the same left operator.

A second differentiation gives

$$\dot{\mathbf{R}}_{,ij} = \mathbf{A}\mathbf{R}_{,ij} + \mathbf{R}_{,ij}\mathbf{A}^\mathsf{T} + \mathbf{Q}_{ij}, \tag{31}$$

with

$$\begin{aligned}\mathbf{Q}_{ij} = {} & \mathbf{A}_{,i}\mathbf{R}_{,j} + \mathbf{R}_{,j}\mathbf{A}_{,i}^\mathsf{T} + \mathbf{A}_{,j}\mathbf{R}_{,i} + \mathbf{R}_{,i}\mathbf{A}_{,j}^\mathsf{T} \\ & + \mathbf{A}_{,ij}\mathbf{R} + \mathbf{R}\mathbf{A}_{,ij}^\mathsf{T} + \mathbf{B}_{,ij}.\end{aligned} \tag{32}$$

The ordering $\mathbf{R} \to \mathbf{R}_{,i} \to \mathbf{R}_{,ij}$ is exact within a stage because the augmented system is lower block triangular.

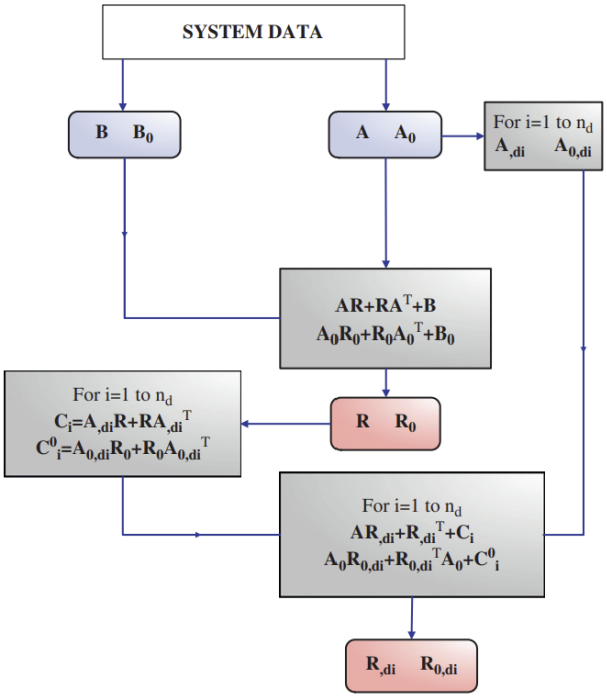


Figure 4: Reference first-order covariance-sensitivity flowchart. The present framework preserves this nominal-to-gradient sequence and appends a Hessian level.

A.3 Joint adaptive integration

The local-error state is the stack

$$\mathcal{X} = (\mathbf{R}, \mathbf{R}_{,1}, \ldots, \mathbf{R}_{,n_p}, \mathbf{R}_{,11}, \mathbf{R}_{,12}, \ldots) \,. \tag{33}$$

Because the blocks have different units, the error norm uses elasticity-like scaling. The first derivatives are scaled by a characteristic $|\theta_i|$, and the second derivatives by $|\theta_i \theta_j|$. A common accepted step is essential: interpolating independently integrated gradient and Hessian histories can create avoidable inconsistency in the moment contractions.

A.4 Transformation to physical response covariance

If

$$\mathbf{C}_y = \mathbf{H}\mathbf{R}\mathbf{H}^{\mathsf{T}}, \tag{34}$$

then the first derivative is

$$\mathbf{C}_{y,i} = \mathbf{H}_{,i}\mathbf{R}\mathbf{H}^{\mathsf{T}} + \mathbf{H}\mathbf{R}_{,i}\mathbf{H}^{\mathsf{T}} + \mathbf{H}\mathbf{R}\mathbf{H}_{,i}^{\mathsf{T}}. \tag{35}$$

The direct terms involving $\mathbf{H}_{,i}$ are required whenever the output definition explicitly depends on a parameter. This occurs, for example, for absolute acceleration when the structural damping ratio is uncertain.

The second derivative is

$$\begin{aligned}\mathbf{C}_{y,ij} = {} & \mathbf{H}_{,ij}\mathbf{R}\mathbf{H}^{\mathsf{T}} + \mathbf{H}_{,i}\mathbf{R}_{,j}\mathbf{H}^{\mathsf{T}} + \mathbf{H}_{,i}\mathbf{R}\mathbf{H}_{,j}^{\mathsf{T}} + \mathbf{H}_{,j}\mathbf{R}_{,i}\mathbf{H}^{\mathsf{T}} \\ & + \mathbf{H}\mathbf{R}_{,ij}\mathbf{H}^{\mathsf{T}} + \mathbf{H}\mathbf{R}_{,i}\mathbf{H}_{,j}^{\mathsf{T}} + \mathbf{H}_{,j}\mathbf{R}\mathbf{H}_{,i}^{\mathsf{T}} + \mathbf{H}\mathbf{R}_{,j}\mathbf{H}_{,i}^{\mathsf{T}} + \mathbf{H}\mathbf{R}\mathbf{H}_{,ij}^{\mathsf{T}}.\end{aligned} \tag{36}$$

Neglecting the derivatives of $\mathbf{H}$ is acceptable only when the output transformation is parameter independent.

Algorithm 1 Joint mixed-order evolutionary-Lyapunov integration.

Require: $\mathbf{A}(\boldsymbol{\theta}, t)$, $\mathbf{B}(\boldsymbol{\theta}, t)$, derivatives $\mathbf{A}_{,i}$, $\mathbf{B}_{,i}$, $\mathbf{A}_{,ij}$, $\mathbf{B}_{,ij}$, initial conditions, tolerances

```
Initialise t, h, R, R_{,i}, and R_{,ij}
while t < t_f do
    Form stage matrices and perform stability diagnostics
    Solve the nominal covariance trial step
    for i = 1, ..., n_p do
        Assemble Q_i and solve the first-sensitivity trial step
    end for
    for i = 1, ..., n_p do
        for j = i, ..., n_p do
            Assemble Q_ij and solve the second-sensitivity trial step
        end for
    end for
    Evaluate a joint scaled local-error norm
    if err ≤ 1 then
        Accept all blocks on the same grid and update h by the PI controller
    else
        Reject all blocks, reduce h, and repeat the trial
    end if
end while
return {t_k, R_k, R_{,i,k}, R_{,ij,k}}
```

A.5 Taylor expansion of scalar and vector outputs

Let $\delta\boldsymbol{\theta} = \boldsymbol{\theta} - \bar{\boldsymbol{\theta}}$ with $\mathbb{E}[\delta\boldsymbol{\theta}] = \mathbf{0}$ and $\mathrm{Cov}[\delta\boldsymbol{\theta}] = \boldsymbol{\Sigma}_\theta$. For a scalar output y,

$$y \approx y_0 + \mathbf{g}_y^\mathsf{T}\delta\boldsymbol{\theta} + \frac{1}{2}\delta\boldsymbol{\theta}^\mathsf{T}\mathbf{H}_y\delta\boldsymbol{\theta}. \tag{37}$$

The first-order approximation retains only $y_0 + \mathbf{g}_y^\mathsf{T}\delta\boldsymbol{\theta}$. For a vector output $\mathbf{q}$, each component has its own gradient and Hessian. In the toolbox, the independent entries of a symmetric matrix are collected through $\mathrm{vech}(\cdot)$.

A.6 First-order second-moment propagation

For output components y_e and y_f,

$$\mu_e^{(1)} = y_{e,0}, \tag{38}$$

$$\mathrm{Cov}^{(1)}[y_e, y_f] = \mathbf{g}_e^\mathsf{T}\boldsymbol{\Sigma}_\theta\mathbf{g}_f. \tag{39}$$

For independent parameters,

$$\mathrm{Var}^{(1)}[y_e] = \sum_{i=1}^{n_p}\left(\frac{\partial y_e}{\partial\theta_i}\right)^2\sigma_{\theta_i}^2, \tag{40}$$

which is the direct perturbation result used in Marano et al. [1]. With correlated parameters, the full double sum is retained.

A.7 Gaussian second-order second-moment propagation

If $\delta\boldsymbol{\theta}$ is zero-mean Gaussian, odd central moments vanish and fourth moments follow Isserlis' theorem. The mixed-order moments become

$$\mu_e^{(2)} = y_{e,0} + \frac{1}{2}\,\mathrm{tr}(\mathbf{H}_e\boldsymbol{\Sigma}_\theta), \tag{41}$$

$$\mathrm{Cov}^{(2)}[y_e, y_f] = \mathbf{g}_e^{\mathsf{T}}\boldsymbol{\Sigma}_\theta\mathbf{g}_f + \frac{1}{2}\,\mathrm{tr}(\mathbf{H}_e\boldsymbol{\Sigma}_\theta\mathbf{H}_f\boldsymbol{\Sigma}_\theta). \tag{42}$$

Equation (42) demonstrates why the method is mixed: the gradient term remains the leading contribution and the Hessian adds a curvature correction. If the parameter distribution is non-Gaussian, the quadratic expansion remains valid locally, but the moment formulas require the parameter third- and fourth-order central-moment tensors [15].

A.8 Application to the covariance matrix

At every accepted time t_k, define

$$\mathbf{q}_k = \mathrm{vech}(\mathbf{C}_y(t_k)). \tag{43}$$

The Jacobian $\mathbf{J}_k$ has rows equal to the gradients of the independent covariance entries. The first-order output covariance is

$$\boldsymbol{\Sigma}_{q,k}^{(1)} = \mathbf{J}_k\boldsymbol{\Sigma}_\theta\mathbf{J}_k^{\mathsf{T}}. \tag{44}$$

The second-order mean is assembled entry by entry using equation (41); the second-order output covariance uses equation (42). The diagonal of $\boldsymbol{\Sigma}_{q,k}$ gives the variance of each calculated covariance entry. This object is stored separately from $\mathbf{C}_y$.

A.9 Derived quantities by the chain rule

For $\sigma_x = \sqrt{C_{xx}}$,

$$\frac{\partial\sigma_x}{\partial\theta_i} = \frac{C_{xx,i}}{2\sqrt{C_{xx}}}, \tag{45}$$

while

$$\frac{\partial^2\sigma_x}{\partial\theta_i\partial\theta_j} = \frac{C_{xx,ij}}{2\sqrt{C_{xx}}} - \frac{C_{xx,i}C_{xx,j}}{4C_{xx}^{3/2}}. \tag{46}$$

Similar chain rules apply to correlation coefficients, spectral ordinates, energy indices, and reliability measures. A square-root transformation of covariance-entry confidence limits is useful for plotting but is not identical to a second-order propagation of σ_x itself.

A.10 Confidence bands and positivity

The moment-matched central band is

$$[\ell_e, u_e] = [\mu_e - z_p\sigma_e, \mu_e + z_p\sigma_e]. \tag{47}$$

For variance entries, ℓ_e is clipped at zero. Element-wise clipping does not preserve joint positive semidefiniteness. The raw SOSM mean and a nearest-PSD projection are therefore both stored. The raw mean is the perturbation result; the projected mean is used only when a valid covariance is required by a multivariate Gaussian density.

A.11 Linear and Bouc–Wen derivative routes

For the linear model, the software evaluates the matrix derivatives and integrates equations (29) and (31). This route preserves the common adaptive grid and is efficient for a full Hessian. For Bouc–Wen equivalent linearisation, $\mathbf{A}_{eq}$ depends on the evolving covariance through a fixed point. The current software computes complete-solver centred finite differences. This includes the implemented nonlinear coupling but is more expensive and requires a perturbation-step study.

A.12 Recommended mixed-order decision logic

A practical sequence is:

1. calculate all first-order sensitivities;
2. rank parameters by their contribution to the output variance;
3. retain influential parameters and calculate the full Hessian for that reduced set;
4. compare the SOSM mean correction with the nominal output;
5. compare the Hessian covariance correction with the gradient covariance term;
6. validate the final moments by sampling if either correction is large.

This strategy avoids spending most of the computation on mixed second derivatives that have negligible influence.

B Robust multiobjective optimization algorithm

This appendix defines an optimization layer that can use SSS-Spectra Adaptive as its analysis kernel. It follows the robust-design philosophy of Marano et al. [1], Zang et al. [16] but uses the mixed first-/second-order moment estimates developed in Appendix A. The natural algorithm is an elitist multiobjective evolutionary procedure such as NSGA-II [17], available in MATLAB through `gamultiobj` or through a dedicated implementation.

B.1 Design variables, uncertain parameters, and performance measures

Let

$$\mathbf{b} = [b_1, \ldots, b_{n_b}]^{\mathsf{T}} \in \Omega_b \tag{48}$$

be the deterministic design vector. Examples include a nominal damping ratio, a nominal natural period, a post-yield stiffness ratio, a yield displacement, or parameters of a control device. Let $\mathbf{d}$ collect uncertain quantities around their nominal values. A design variable may also define the mean of an uncertain parameter; in that case its coefficient of variation or standard deviation must be updated consistently.

Let $J(\mathbf{b}, \mathbf{d})$ be a scalar performance measure. Typical choices are:

- maximum displacement variance or maximum displacement standard deviation;
- probabilistic peak S_d, S_v, or S_a;
- probability of failure or negative reliability;
- maximum hysteretic-energy demand;
- a weighted combination of response, energy, cost, or material quantities.

The robust objective vector is

$$\mathbf{F}(\mathbf{b}) = \begin{bmatrix} \mu_J(\mathbf{b}) \\ \sigma_J(\mathbf{b}) \end{bmatrix}. \tag{49}$$

The first objective controls expected performance. The second controls sensitivity to uncertainty and therefore robustness.

B.2 Mixed-order objective evaluation

At every candidate design, SSS-Spectra Adaptive calculates the nominal performance J_0, its gradient $\mathbf{g}_J$, and, when requested, its Hessian $\mathbf{H}_J$. The mixed-order estimates are

$$\mu_J \approx J_0 + \frac{1}{2}\,\mathrm{tr}(\mathbf{H}_J \boldsymbol{\Sigma}_d), \tag{50}$$

$$\sigma_J \approx \left[\mathbf{g}_J^{\mathsf{T}} \boldsymbol{\Sigma}_d \mathbf{g}_J + \frac{1}{2}\,\mathrm{tr}(\mathbf{H}_J \boldsymbol{\Sigma}_d \mathbf{H}_J \boldsymbol{\Sigma}_d)\right]^{1/2}. \tag{51}$$

A first-order optimization is obtained by dropping the Hessian terms. A cost-effective mixed strategy evaluates the entire population with FOSM, identifies the current non-dominated region, and applies the complete SOSM calculation only to elite candidates or to the final Pareto set.

B.3 Constraints

Deterministic constraints can be written as $g_k(\mathbf{b}) \le 0$. A moment-based robust constraint is

$$\mu_{G_k}(\mathbf{b}) + \kappa_k \sigma_{G_k}(\mathbf{b}) - G_{k,\mathrm{adm}} \le 0, \tag{52}$$

where κ_k sets a one-sided safety margin. Reliability constraints may alternatively take the form

$$P_f(\mathbf{b}) \leq P_{f,\mathrm{adm}}. \tag{53}$$

When reliability is calculated from the excitation-induced crossing process but uncertain parameters are also present, the reliability should be differentiated or sampled consistently; treating a nominal P_f as a robust constraint is not sufficient.

B.4 NSGA-II operations

NSGA-II maintains a population of candidate designs and ranks them by Pareto dominance. Candidate a dominates candidate b when it is no worse in every objective and strictly better in at least one. Within the same non-domination rank, crowding distance favours solutions in sparsely populated objective-space regions and preserves front diversity.

The standard operations are:

1. random or space-filling population initialisation inside the bounds;
2. candidate analysis and robust-objective evaluation;
3. non-dominated sorting and crowding-distance calculation;
4. binary tournament selection using rank and crowding distance;
5. simulated binary crossover for real-valued variables [18];
6. polynomial mutation;
7. elitist combination of parent and offspring populations;
8. truncation to the prescribed population size;
9. termination after a generation limit or after Pareto-front stabilisation.

B.5 Recommended two-stage computational strategy

A full Hessian at every candidate can dominate the cost of an evolutionary search. The following two-stage strategy is recommended.

B.5.1 Stage 1: first-order global search

Use all uncertain parameters but only nominal covariance and gradients. Run a broad NSGA-II search with moderate integration tolerance. The objectives are J_0 and $\sqrt{\mathbf{g}_J^{\mathsf{T}} \boldsymbol{\Sigma}_d \mathbf{g}_J}$. Record parameter importance and eliminate consistently negligible uncertainty sources.

B.5.2 Stage 2: second-order Pareto refinement

Restart from the non-dominated Stage-1 designs. Use the reduced uncertain-parameter set, full mixed Hessians, tighter tolerances, and the SOSM objectives in equation (51). Apply local perturbations or a smaller evolutionary population around the Stage-1 front. Finally, validate selected knee points by Monte Carlo or stochastic collocation.

This strategy is not an approximation of the final selected designs: the reported final Pareto set is second-order. The first-order stage is a computational screening mechanism.

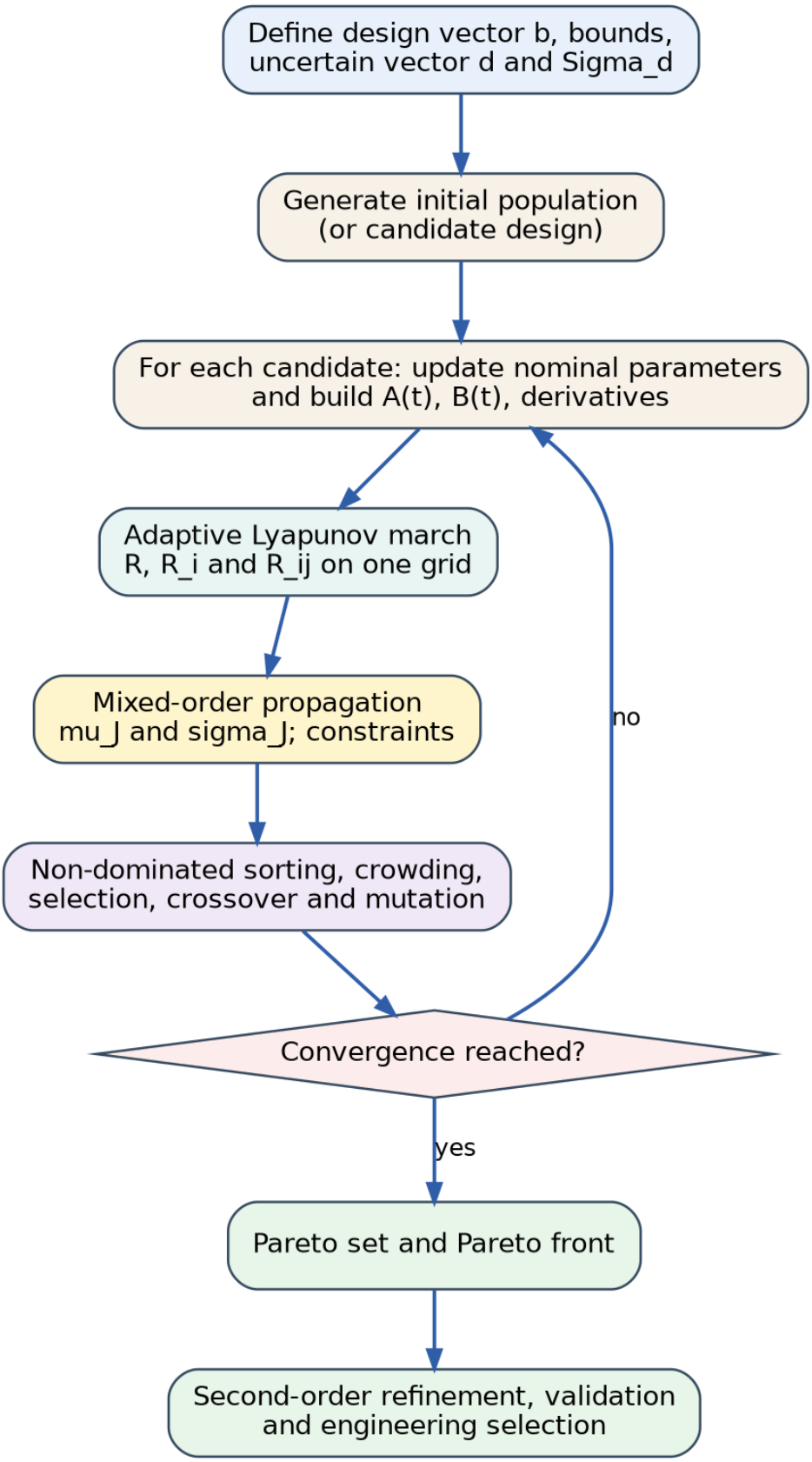


Figure 5: Robust multiobjective optimization workflow using SSS-Spectra Adaptive as the candidate-evaluation kernel.

Algorithm 2 Robust NSGA-II optimization with mixed-order stochastic analysis.

Require: Design bounds, parameter covariance $\mathbf{\Sigma}_d$, population size N, maximum generations N_g

Generate initial population $\mathcal{P}_0$
for each candidate $\mathbf{b} \in \mathcal{P}_0$ **do**
    Update nominal model parameters and uncertainty data
    Run nominal and first-order covariance-sensitivity analysis
    Compute FOSM objectives and constraints
    if full mixed-order evaluation is enabled **then**
        Compute Hessians and SOSM objective corrections
    end if
end for
for $g = 0, \ldots, N_g - 1$ **do**
    Non-dominated sort and calculate crowding distances
    Create mating pool by binary tournaments
    Generate offspring by simulated binary crossover and polynomial mutation
    for each offspring candidate **do**
        Evaluate robust objectives and constraints through SSS-Spectra Adaptive
    end for
    Combine parents and offspring, sort, and retain the best N candidates
    if front convergence criterion is satisfied **then**
        break
    end if
end for
Re-evaluate the final Pareto set with full second-order moments and tighter tolerances
return Pareto designs, robust objectives, constraints, and diagnostics

B.6 MATLAB-oriented objective wrapper

The following listing illustrates a generic objective function. The design variables are mapped into `P`; the uncertain analysis is run; and the maximum over time of the displacement-covariance mean and standard deviation is returned. A different response metric can be substituted without changing the optimizer.

```
function f = robust_objectives(b,P0,model,Tn,designNames,useSecondOrder)
    P = P0;
    for k = 1:numel(designNames)
        P = sss_setparam(P,designNames{k},b(k));
    end

    if useSecondOrder
        P.unc.order = 2;
        branch = 'second_order';
    else
        P.unc.order = 1;
        branch = 'first_order';
    end

    OUT = sss_run_uncertain_case(P,model,Tn);
    S = OUT.uncertainty.response_covariance.(branch);

    muCxx = squeeze(S.mean(1,1,:));
    sdCxx = squeeze(S.std(1,1,:));

    f(1) = max(muCxx); % expected displacement variance
    f(2) = max(sdCxx); % dispersion of calculated variance
```

```
end
```

A MATLAB `gamultiobj` call can then be written as

```
obj = @(b) robust_objectives(b,P,'linear',0.50, ...
                             {'xi','wg'},false);
opts = optimoptions('gamultiobj', ...
    'PopulationSize',120, ...
    'MaxGenerations',150, ...
    'UseParallel',true, ...
    'Display','iter');
[bPareto,fPareto] = gamultiobj(obj,2,[],[],[],[],lb,ub,[],opts);
```

After Stage 1, every Pareto candidate should be re-evaluated with `useSecondOrder = true`. If a response standard deviation rather than a covariance entry is optimized, calculate its gradient and Hessian by the chain rule instead of merely taking the square root of covariance-entry bands.

B.7 Population and operator settings

Table 5: Indicative NSGA-II settings for an expensive stochastic analysis.

Setting	Recommended starting point
Population size	60–150, increasing with the number of design variables.
Generations	80–200 for Stage 1; fewer for second-order refinement.
Crossover probability	0.85–0.95.
Mutation probability	Approximately $1/n_b$ per variable.
Parallel execution	Enable when independent candidate evaluations can run on separate workers.
Caching	Cache designs after rounding to a tolerance to avoid duplicate expensive analyses.
Integration tolerance	10^{-4} to 10^{-5} in Stage 1; tighten for final validation.
Hessian mode	Diagonal for screening; full for final mixed-order Pareto designs.

B.8 Pareto-front interpretation and design selection

The optimizer returns a set, not a unique solution. Selection should be based on engineering priorities. A knee point provides a large reduction in uncertainty for a limited deterioration of expected performance. Objective values should be normalised before distance- or curvature-based knee detection. Feasibility, constructability, parameter bounds, and model-validity limits must be applied before selecting a design. A solution with a slightly worse mean but a substantially smaller standard deviation may be preferable because it is less sensitive to imperfect parameter knowledge.

B.9 Validation of optimal designs

At least three designs should be validated independently: the minimum-mean extreme, the minimum-dispersion extreme, and the selected compromise. Validation should include tighter adaptive tolerances, finite-difference step checks, full Hessians, and a sampling method. If the empirical mean or variance differs significantly from the SOSM prediction, the local Taylor model is not adequate in that region and the optimization must use a more global uncertainty method.

B.10 Extension to reliability-based and energy-based design

The same algorithm applies when J is a failure probability or an energy measure. For reliability, differentiate the out-crossing rate and hazard integral or evaluate reliability for perturbed parameter samples. For energy, propagate the gradient and Hessian of the chosen cumulative or peak energy quantity. Multiple robust objectives can be included, for example expected peak displacement, its standard deviation, expected hysteretic energy, and material cost. NSGA-II remains applicable, but high-dimensional Pareto fronts require careful visualisation and decision support [19].

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